
Research Interests

My research interests lie at the intersection of cryptography, distributed systems, and machine learning, with a focus on secure multi-party computation and privacy-preserving distributed and machine learning systems. I am broadly interested in combining formal security guarantees with practical decentralized and ML systems to enable secure deployment of modern computing technologies.

Education

- 2018–2024 **Ph.D. in Computer Science**, *Carnegie Mellon University*, Pittsburgh, USA
Thesis: Distributed Cryptography as a Service
Advisors: Prof. Bryan Parno, Prof. Vipul Goyal.
Selected graduate coursework: Graduate Artificial Intelligence, Foundations of Distributed Consensus and Blockchains, Graduate Algorithms
- 2016–2019 **M.Sc. in Computer Science**, *Karlsruhe Institute of Technology (KIT)*, Germany
Focus: Cryptography and algorithms
Master's thesis: Rerandomizing Oblivious Transfer for Online Oblivious Bingo Voting
Advisors: Prof. Jörn Müller-Quade, Dr. Rebecca Schwerdt
- 2013–2018 **B.Sc. in Mathematics**, *KIT*, Karlsruhe, Germany
- 2011–2016 **B.Sc. in Computer Science**, *KIT*, Karlsruhe, Germany
Undergraduate thesis (conducted at Carnegie Mellon University): Evaluation of Publishable Humanly Usable Secure Password Creation Schemas
Advisors: Prof. Manuel Blum, Prof. Jörn Müller-Quade

Work Experience

- 2024–present **Carnegie Bosch Postdoctoral Fellow**, *Carnegie Mellon University*, USA
Host: Prof. Elaine Shi. Projects include privacy-preserving machine learning utilizing differential privacy, secure multi-party computation, and LLM techniques.
- 2024 **Summer Research Associate**, *JP Morgan Chase AI Research*, NYC, USA
Developed and implemented zero-knowledge proof systems for verifiable auditing of machine learning models, including protocol design, prototype implementation, and empirical evaluation on real-world datasets.
- 2023 **Summer Research Associate**, *JP Morgan Chase AI Research*, NYC, USA
Designed secure multi-party computation protocol for large-scale distributed systems based on packed secret-sharing.
- 2019, 2020 **Teaching Assistant**, *Carnegie Mellon University*, Pittsburgh, USA
- 2014–2018 **Freelancing in Software- and Web-Development**, Karlsruhe, Germany
Developed and deployed custom full-stack web applications and dashboards for startups, non-profits, and public sector clients, supporting user management and operational workflows.
- 2014–2018 **Student Research Assistant**, *Karlsruhe Institute of Technology*, Germany
Implemented educational tools such as visualizations of zero-knowledge principles, contributed to research related to CCA-secure commitments, private set intersection, Intel SGX.
- 2013–2014 **Student Software Developer**, *bluehands GmbH & Co.munication KG*, Germany
Developed software components for an augmented reality mobile application.
- 2013 **Student Research Assistant**, *FZI Research Center for Information Technology*
Developed a motion sensor-based mobile rehabilitation app for stroke patients.

Honors, Grants and Awards

- 2024 **Carnegie Bosch Postdoctoral Fellowship**, Pittsburgh, USA
Two-year award for postdoctoral researchers conducting high-potential research supported by the CBI institute (collaboration of Bosch and CMU)

- 2022 **Protocol Labs Research Gift**, 35,000 USD, Pittsburgh, USA
Project: Stateless Distributed Randomness Generation
- 2015 **InterAct scholarship**, Karlsruhe, Germany
Scholarship awarded to conduct undergraduate thesis research in the USA
- 2014 **Microsoft Imagine Cup**, Berlin, Germany
Winner of the national final (team “Krowd”), Innovation category
Project: System to monitor crowd movement by tracking signals of WiFi enabled devices to ensure security at large-scale events
- 2014 **Winner of HackZurich**, Zurich, Switzerland
Winner of Europe’s largest hackathon (team “Immersive”)
- 2012-2014 **ACM ICPC**, Netherlands and Germany
German Collegiate Programming Contest (GCPC),
Northwestern European Regional Contest (NWERC)

Publications & Under Review

Unless noted otherwise, author order is **alphabetical**.

Security and Privacy for ML

- USenix ’26, RegML @NeurIPS ’25 Certified in Theory, Broken in Practice: Assumption Gaps in Cryptographic Model Certification. Carter Luck[†], Olive Franzese[†], Elisaweta Masserova[†], Akira Takahashi, Antigoni Polychroniadou, Nicolas Papernot. [†]Equal contribution. *USENIX Security Symposium (USENIX-Security) 2026. Further accepted at the Regulatable ML Workshop at Annual Conference on Neural Information Processing Systems (NeurIPS)*.
- UAI ’26 AudiFair: Privacy-Preserving Framework for Auditing Fairness. Elisaweta Masserova, Kshitij Kulkarni, Antigoni Polychroniadou, Ron D. Rothblum, Akira Takahashi. Author order reflects contribution. *The Conference on Uncertainty in Artificial Intelligence (UAI) 2026*.
- RegML @NeurIPS ’24 Non-Interactive and Publicly Verifiable Zero-Knowledge Proof for Fair Decision Trees. Elisaweta Masserova, Antigoni Polychroniadou, Akira Takahashi. *Regulatable ML Workshop at Annual Conference on Neural Information Processing Systems (NeurIPS) 2024*. Selected for **oral presentation**.

Cryptography for (Large-Scale) Distributed Systems

- SODA ’27 (in submission) On the Impossibility of Perfectly Secure Coin-Toss for $n < 3t+1$. Elisaweta Masserova. *In submission to the ACM-SIAM Symposium on Discrete Algorithms (SODA) 2027*.
- Eurocrypt ’27 (in submission), TPMPC ’26 Asynchronous MPC over Layered Graphs without Trusted Setup. Gabriel Dettling, Chen-Da Liu-Zhang, Elisaweta Masserova. *In submission to the Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT) 2027. Further accepted at Theory and Practice of Multi-Party Computation Workshop (TPMPC)*.
- CRYPTO ’26 Broadcast for Dynamic Committees without Trusted Setup. Gabriel Dettling, Chen-Da Liu-Zhang, Elisaweta Masserova, Matthieu Rabaud, Antoine Urban. *Annual International Cryptology Conference (CRYPTO) 2026*.
- EUROCRYPT ’25, TPMPC ’25 Efficient Distributed Randomness Generation from Minimal Assumptions where Parties Speak Sequentially Once. Chen-Da Liu-Zhang, Elisaweta Masserova, João Ribeiro, Pratik Soni, Sri Aravinda Krishnan Thyagarajan. *Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT) 2025. Further accepted at Theory and Practice of Multi-Party Computation Workshop (TPMPC)*.
- ACM PODC ’25 Brief Announcement: Towards Scalable YOSO MPC via Packed Secret-Sharing. Daniel Escudero, Elisaweta Masserova, Antigoni Polychroniadou. *44th ACM Symposium on Principles of Distributed Computing (PODC) 2025*.

- ASIACRYPT '25 Towards Scalable YOSO MPC via Packed Secret-Sharing. Daniel Escudero, Elisaweta Masserova, Antigoni Polychroniadou. *Annual International Conference on the Theory and Applications of Cryptology and Information Security (ASIACRYPT)* 2025.
- FC '24, Improved YOSO Randomness Generation with Worst-Case Corruptions. Chen-Da
CTB '24 Liu-Zhang, Elisaweta Masserova, João Ribeiro, Pratik Soni, and Sri AravindaKrishnan Thyagarajan. *International Conference on Financial Cryptography and Data Security (FC)* 2024. *Further accepted as talk at Eurocrypt's Workshop on Cryptographic Tools for Blockchains (CTB)* 2024.
- PKC '22 Storing and Retrieving Secrets on a Blockchain. Vipul Goyal, Abhiram Kothapalli, Elisaweta Masserova, Bryan Parno, Yifan Song. *International Conference on Practice and Theory in Public Key Cryptography (PKC)*, 2022.
- TCC '21 Blockchains Enable Non-Interactive MPC. Vipul Goyal, Elisaweta Masserova, Bryan Parno, Yifan Song. *Theory of Cryptography Conference (TCC)*, 2021.
- ACSAC '20 Talek: Private Group Messaging with Hidden Access Patterns. Raymond Cheng, William Scott, Elisaweta Masserova, Irene Zhang, Vipul Goyal, Thomas Anderson, Arvind Krishnamurthy, Bryan Parno. *Annual Computer Security Applications Conference (ACSAC)*, 2020.

Game Theory and Mechanism Design

- DISC '26 Fairness in the Wild: Secure Atomic Swap with External Incentives. Hao Chung, Elisaweta Masserova, Elaine Shi, Sri AravindaKrishnan Thyagaraja. *International Symposium on Distributed Computing (DISC)*, 2026.
- EUROCRYPT '26 Game Theory Does Not Always Help: The Case of Statistical Multi-Party Coin Tossing. Chen-Da Liu-Zhang, Elisaweta Masserova, João Ribeiro, Sri AravindaKrishnan Thyagarajan. *Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT)* 2026.
- FC '25, Rapidash: Atomic Swaps Secure under User-Miner Collusion. Hao Chung, Elisaweta
SBC '24 Masserova, Elaine Shi, Sri AravindaKrishnan Thyagarajan. *International Conference on Financial Cryptography and Data Security (FC)* 2025. *Further accepted as talk at the Science of Blockchain Conference (SBC)* 2024.
- CESC '22 Ponyta: Foundations of Side-Contract-Resilient Fair Exchange. Hao Chung, Elisaweta Masserova, Elaine Shi, Sri AravindaKrishnan Thyagaraja. *Crypto Economics Security Conference (CESC)*, 2022.

Manuscripts

- 2022 Logic Locking via Universal Circuits On a Grid Topology. Deepali Garg, Vipul Goyal, Ken Mai, Elisaweta Masserova, Bryan Parno, and Lawrence Pileggi.
- 2020 Poppins: A Direct Construction for Asymptotically Optimal zkSNARKs. Abhiram Kothapalli, Elisaweta Masserova, Bryan Parno.

Selected Invited Talks

- Apr 2025 Introduction to Cryptography (invited lecture). *Polkadot Blockchain Academy, PBA-X*, online course, over 300 participants.
- Apr 2025 MEV and its Countermeasures. *Workshop on Recent Advances in Fairness in Distributed Applications*, Miyakojima, Japan.
- Oct 2024 Efficient Distributed Randomness Generation from Minimal Assumptions in PASSO model. *Security and Privacy seminar at UPenn*, Philadelphia, USA.
- May 2023 Storing and Retrieving Secrets on a Blockchain. *Secure Blockchain Summit at CMU*, Pittsburgh, USA.

Academic Service

Organizer Roles

2025 Committee Member of CMU's Secure Blockchain Summit
2021-2024 Organizer of The CMU CyLab Crypto Seminar

Program Committee

2026 IEEE S&P, FC
2025 USENIX, ACM CCS, WWW, FC, LATINCRYPT, CVC
2024 ACM CCS, FC

Technical Skills

Languages Python, TypeScript, Solidity, C#, C++, Rust, PHP
Cryptography MPC frameworks, zkSNARKS, implementation using arkworks
ML Scikit-learn, PyTorch
Web React, Next.js, Node.js
Systems & Tools Linux, Git